

Mukesh Patel School of Technology Management & Engineering
School of Technology Management and Engineering

Course Policy Document

Course Name - (Code): Cyber Forensic -702IT0C019

Program and Semester: B. Tech (IT) – Semester VI	Pre-requisite Course: Basic Knowledge of Computer Network, Operating Systems and programming					
Academic Year: 2024-25	Credit Details:	L	T	P	C	H
		2	0	2	3	4
Name of Course Faculty: Prof Pintu Shah Program: B. Tech (IT) , MBA(Tech) IT and BTI(EXTC) Contact Details: +91 9820957833 Email: pintu.shah@nmims.edu Office Hours:	Faculty associated with the course: - Prof. Mohnish Singh (MPSTME, Mumbai campus) Program: B.tech(Tech) - Computer Engineering Contact Details: +91 9930312027 Email: singhmohnishpadam@gmail.com Office Hours: As per lecture timings (Visting faculty)					
Pre-Course Activity: Read through basics of Windows and Linux operating system and computer network.						
Course link:						

1. Introduction to the Course

1.1 Importance of the Course

Dive into the fascinating world of digital forensics and incident response with this comprehensive introductory course. Designed for students eager to explore the intricacies of cybercrime, this course covers a wide range of topics, equipping you with the essential skills and knowledge to perform forensic operations on various digital media

Computer forensics, also called digital or cyber forensics, is a technology field that uses investigation techniques to help identify, collect, and store evidence from an electronic device. As the world becomes more connected digitally, digital evidence for solving crimes is becoming more relevant daily. A computer forensics investigator's job is to collect, examine, and safeguard this evidence to help solve cybercrimes and to recover important compromised data.

This course is an introduction to the field of cyber forensics. This course introduces various aspects of cyber forensics to the students. Students will learn about various methods, tools and techniques to perform forensic operations on a given media.

Definition of cybercrime, type and classification of cybercrimes, Overview of cyber forensics, forensic process, types of investigations, digital evidence, data acquisition

and forensic imaging, digital forensic tools

This course is perfect for anyone interested in cybersecurity careers or enhancing their knowledge in digital forensics. With a blend of theoretical knowledge and practical skills, you will be well-prepared to tackle real-world challenges in the realm of cybercrime investigation and incident management. Join us to unlock your potential in this dynamic field

1.2 Objectives of the Course:

This course is an introduction to the field of digital forensics and incident response. This course introduces various aspects of digital forensics and incident response to the students. Students will learn about various methods, tools and techniques to perform forensic operations on a given media.

1. Introduction to the field of cyber forensics and cybercrimes.
2. Introduction to various aspects of cyber forensics.
3. Learn various methods, tools and techniques to perform forensic operations on a given media.
4. Understand various legal, professional and ethical issues involved in cyber forensics.

2. Course Outcomes (CO), Mapping with Program Outcomes (PO), and Program Specific Outcomes (PSO)

2.1 Course Outcomes

- 2.1.1 CO1:** Describe Windows and Linux File systems
- 2.1.2 CO2:** Perform various forensics operations on a given media
- 2.1.3 CO3:** Describe various professional, ethical and legal issues related to cyber forensics

2.2 Program Outcomes(PO) the course contributed to:

- 2.2.1 PO-1:** An ability to apply knowledge of mathematics, science, and engineering for problem-solving.
- 2.2.2 PO-2:** An ability to research, design and conduct experiments, as well as to analyze and interpret data.
- 2.2.3 PO-3:** An ability to design, implement, and evaluate a computer-based system, process, component, or program to meet desired needs.
- 2.2.4 PO-4:** An ability to function effectively on teams to accomplish a common goal.

- 2.2.5 PO-5:** An ability to identify, formulate and provide effective IT solutions for engineering problems.
- 2.2.6 PO-6:** An understanding of professional, legal, security and social issues and responsibilities.
- 2.2.7 PO-7:** An ability to communicate effectively with a range of audiences.
- 2.2.8 PO-8:** The broad education necessary to understand the impact of engineering solutions in a global, economic, environmental, and societal context.
- 2.2.9 PO-9:** Recognition of the need for and an ability to engage in continuing professional development and self-learning.
- 2.2.10 PO-10:** An ability to apply ethical principles in the **development** of IT solutions.
- 2.2.11 PO-11:** An ability to use the techniques, skills, and modern engineering tools necessary for developing effective IT solutions.
- 2.2.12 PO-12:** An ability to identify and analyze user needs and take them into account in the selection, creation/integration, evaluation and administration of IT-based solutions.

2.3 Program Specific Outcomes (PSO):

- 2.3.1 PSO-1:** Demonstrate an ability to visualize, architect and create appropriate solutions for IT-related projects.
- 2.3.2 PSO-2:** Demonstrate an ability to professionally manage, monitor and safeguard IT resources.

CO-PO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1		1									2			
CO2	2							3	1		1	2		2
CO3						1	2	3				2		

Mapping Levels: 1- High, 2-Medium, 3-Low

2.4 Student Outcomes (SO) (For ABET accredited Programs):

- SO 1:** Analyze a complex computing problem and to apply principles of computing and other relevant disciplines to identify solutions.
- SO 2:** Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline.
- SO 3:** Communicate effectively in a variety of professional contexts.

SO 4: Recognize professional responsibilities and make informed judgments in computing practice based on legal and ethical principles.

SO 5: Function effectively as a member or leader of a team engaged in activities appropriate to the program's discipline.

SO 6: Identify and analyze user needs and to consider them in the selection, creation, integration, evaluation, and administration of computing-based systems.

3. Teaching-learning methodology

3.1 Instruction Plan

Lecture No.	Topic	Teaching Method *	Blooms Level	Resources	COs mapped	Assessment and Evaluation
1	Introduction to the course	Presentations	Understand	PPT		
2	Definition of cybercrime, type and classification of cybercrimes	Presentations/White board/ real world scenarios	Understand	PPT, TB1	CO3	MT1 & Lab, TEE
3	Overview of cyber forensics and forensic process	Presentations/ Whiteboard	Understand	PPT, TB1	CO2	MT1 & Lab, TEE
4	Types of investigations, digital evidence, data acquisition and forensic imaging,	Presentations/ Whiteboard	Understand	PPT, TB1	CO2	MT1, TEE
5	Digital forensic tools	Presentations/ Hands-on	Apply	PPT, TB1	CO2	MT1 & Lab, TEE
6	Incident Response	Presentations/ Hands-on	Understand	PPT, TB1	CO2	MT1, TEE
7	File system analysis: Understanding file system, booting process, disk drives and solid state devices	Presentations	Understand	PPT, TB1	CO1	MT1, TEE
8	FAT and NTFS file systems	Presentations	Understand	PPT, TB1	CO1	MT1, TEE
9	FAT and NTFS file systems (Contd.)	Presentations/ Hands-on	Apply	PPT, TB1	CO1	MT1, Project & Lab , TEE

10	Partition table and MFT, understanding whole disk encryption	Presentations/ Hands-on	Apply	PPT, TB1	CO1	MT1, Project & Lab, TEE
11	Window registry File structure of ext4.	Presentations/ Whiteboard	Analyze	PPT, TB1	CO1	MT1, Project & Lab, TEE
12	Class Test 1					
13	Data analysis: Preparation for forensic analysis	Presentations	Understanding	PPT, TB1	CO2	TEE
14	Data carving, recovering graphics files	Presentations/ Hands-on	Apply	PPT, TB1	CO2	MT2, Project & Lab, TEE
15	Header analysis,	Presentations/ Whiteboard	Analyze	PPT, TB1	CO2	MT2, Project & Lab, TEE
16	Hexcode analysis	Presentations/ Whiteboard	Analyze	PPT, TB1	CO2	MT2, Project & Lab, TEE
17	Linux, Machintosh file structures	Presentations	Understanding	PPT, TB1	CO2	TEE
18	Linux forensic tools	Presentations/ Hands-on	Apply	PPT, TB1	CO2	MT2, Project & Lab, TEE
19	email and internet activity analysis,	Presentations/ Whiteboard	Analyze	PPT, TB1	CO2	MT2, Project & Lab, TEE
20	Graphic file formats data compressions	Presentations	Understanding	PPT, TB1	CO2	TEE
21	recover graphics files identify unknown file formats Explain copyright issues with graphics	Presentations/ Whiteboard	Analyze	PPT, TB1	CO2	MT2, Project & Lab, TEE
22	Data hiding techniques.	Presentations	Understanding	PPT, TB1	CO2	MT2, Project & Lab, TEE
23	Malware analysis.	Presentations/ Whiteboard	Analyze	PPT, TB1	CO2	MT2 & Lab, TEE
24	Network Forensics Overview	Presentations	Understanding	PPT, TB1	CO2	MT2 & Lab, TEE
25	Analyzing network traffic	Presentations/ Whiteboard	Analyze	PPT, TB1	CO2	MT2 & Lab, TEE

26	Network-based evidence	Presentations	Analyze	PPT, TB1	CO2	MT2 & Lab, TEE
27	Investigating routers.	Presentations	Analyze	PPT, TB1	CO2	MT2, TEE
28	Analyzing DHCP	Presentations/ Whiteboard	Analyze	PPT, TB1	CO1	MT1, Project & Lab, TEE
29	DHCP snooping attack	Presentations/ Whiteboard	Analyze	PPT, TB1	CO1	MT1, Project & Lab, TEE
30	Defense in depth	Presentations	Understanding	PPT, TB1	CO2	TEE
31	Mobile Device Forensics Mobile phone basics, Inside mobile devices	Presentations	Understanding	PPT, TB1	CO2	Lab, TEE
32	data to analyze in a digital forensics investigation Tools used to validate data Explain common	Presentations	Understanding	PPT, TB1	CO2	Lab, TEE
33	standard procedures for conducting forensic analysis of virtual machines	Presentations	Understanding	PPT, TB1	CO2	Lab, TEE
34	Describe the process of a live acquisition	Presentations	Understanding	PPT, TB1	CO2	Lab, TEE
35	Cyber Incidents and response	Presentations	Understanding	PPT, TB1	CO2	Lab, TEE
36	Incidents response Lifecycle and types	Presentations	Understanding	PPT, TB1	CO2	Lab, TEE
37	Class Test 2					
38	Acquisition procedure for mobile devices, mobile forensic tools.	Presentations/ Hands-on	Apply	PPT, TB1	CO2	Lab, TEE
39	data-hiding techniques	Presentations	Understanding	PPT, TB1	CO3	MT2 and Lab, TEE
40	Legal, Professional and Ethical issues Cyber laws in India	Presentations	Understanding	PPT	CO3	MT2 and Lab, TEE
41	Various ethical dilemma and professional conduct,	Presentations	Understanding	PPT, TB1	CO3	MT2 and Lab, TEE

42	Report writing	Presentations	Analyze	PPT, TB1	CO3	Project & Lab, TEE
43	Case Study 1	Presentations			CO1,2,3	
44	Case Study 2	Presentations			CO1,2,3	
45	Course Wrap Up					

TB 1: B. Nelson, A. Philips and C. Steuart, *Guide to Computer Forensics and Investigations*, 6th Edition, Cengage, 2019.

TB 2: N. Jain and D. R. Kalbande, *Digital Forensic: The Fascinating World of Digital Evidences*, Wiley India, 2019.

4. Assessment and Evaluation Scheme

	Internal Continuous Assessment (ICA) (50 marks)					
	Class Tests (20 Marks)		Term-work (30 Marks)			
Assessment Component	Class Test-1	Class Test-2	Lab Submissions	Group Project	Research Paper Presentation	Class participation
Marks	10	10	10	10	05	05

4.1 Internal Continuous Assessment (ICA) – 50 marks

1. Class test 1 and 2

- Two class tests will be conducted as per the academic calendar.
- It will be conducted offline for 10 marks each.
- Class Test 1 will cover unit 1 to unit 4 Class test will cover unit 4 to Unit 8

2. Lab performance evaluation (10 marks)

Each lab will be assessed for 10 marks. Following points will be considered while assessment

- Lab implementation and troubleshooting (6 marks)
- Documentation and timely submission (4 marks)

Note: **50% penalty for late submission.**

3. Group Project (10 marks)

- Students will form a group of 2-3 students. This activity should be completed in the first week.
- Each group will analyze a given forensic image and write a forensic investigation report. Report template will be shared by the faculty.
- Report and presentation submission deadline is **March 31, 2025.**
- Marks distribution:
 - Appropriate analysis – 5 marks
 - Appropriate report – 5 marks

Project Overview

In this group project, students will simulate a real-world cyber incident response scenario. The objective is to apply the concepts learned throughout the course, including digital forensics techniques, incident response strategies, and ethical considerations, to investigate and respond to a simulated cybercrime incident.

Project Objectives

- To collaboratively analyze a cyber incident from detection to resolution.
- To apply forensic methods to gather and analyze digital evidence.
- To develop an incident response plan and execute it in a simulated environment.
- To enhance teamwork and communication skills while addressing legal and ethical issues in digital forensics.

Project Components

1. Scenario Development

- Each group will create a detailed scenario involving a cyber incident (e.g., ransomware attack, data breach, insider threat).
- The scenario should include background information, potential indicators of compromise (IoCs), and the impact on the organization.

2. Incident Response Plan

- Develop a comprehensive incident response plan based on the created scenario. This should include:
 - Preparation: Define roles and responsibilities within the team.
 - Detection: Identify how the incident was detected and what initial evidence was collected.
 - Analysis: Outline methods for analyzing the evidence (e.g., file system analysis, network traffic analysis).
 - Containment: Describe steps taken to contain the incident and prevent further damage.
 - Eradication: Detail how the threat was removed from the environment.
 - Recovery: Explain how systems were restored and secured after the incident.
 - Lessons Learned: Discuss what could be improved in future responses.

3. Forensic Analysis

- Each group will conduct a mock forensic analysis of provided digital evidence related to their scenario. This could include:
 - Analyzing disk images for deleted files or malware.
 - Examining logs for suspicious activity.
 - Performing network traffic analysis to identify malicious communications.

4. Presentation

- Prepare a presentation summarizing the scenario, incident response plan, forensic analysis findings, and lessons learned. Each group will present their project to the class, simulating a briefing to stakeholders.

Deliverables

- A written report that includes:
 - Scenario description
 - Incident response plan
 - Forensic analysis results
 - Ethical considerations and legal implications
- A presentation (15-20 minutes) summarizing key aspects of the project.

Evaluation Criteria

- Clarity and creativity of the scenario (20%)
- Completeness and practicality of the incident response plan (30%)
- Depth of forensic analysis conducted (30%)
- Quality of presentation (20%)

- 4. Class Participation (05 marks)** - The faculty will ask questions during the class or may give some in-class assignment/quiz. Successful response to the question or completion of the in-class activity will fetch marks for the students. The idea

is to encourage students to participate actively in the class. Your class attendance will be considered as a part of class participation.

5. Research Paper presentation (05 marks)

- a. Students will form a group of 2-3 students. This activity should be completed in the first week.
- b. Each group will read at-least read two recent (post 2020) research paper from reputed journals or conferences from the cyber forensics domain.
- c. Students will give presentation of the research paper as per the schedule.
- d. Following points will be considered for grading
 - i. Timeliness of submission
 - ii. Presentation quality
 - iii. Understanding of research
 - iv. QnA

Example research topics

1. The Role of Digital Forensics in Cybercrime Investigations: Analyzing how digital forensics techniques are applied in various types of cybercrime cases.
2. File System Analysis Techniques: A comparative study of FAT and NTFS file systems in the context of digital forensics.
3. Data Carving Methods in Digital Forensics: Evaluating different data recovery techniques and their effectiveness in forensic investigations.
4. Network Forensics: Analyzing Network Traffic for Evidence: Exploring methodologies for capturing and analyzing network-based evidence during an incident response.
5. Mobile Device Forensics: Challenges and methodologies for acquiring evidence from mobile devices in criminal investigations.
6. Dark Web Forensics: Investigating the unique challenges and techniques used to gather evidence from the dark web.

4.2 Term End Examination (TEE) – (100 marks scaled down to 50)

TEE will be conducted at the end of the semester will be for 100 marks (3-hour duration). Marks obtained scaled down to 50. There will be 7 questions (each question will be 20 marks). Q1 will be compulsory and any 4 from the remaining to be solved.

4.3 Course Passing Criteria

4.3.1 ICA (50 marks) – No minimum marks

4.3.2 TEE (100 marks scaled to 50) – 40% required for passing

4.4 Assessments and Mapping to Course Outcomes

	Internal Continuous Assessment (ICA)					Term End Examination (TEE)
Course Outcomes	CT-1	CT-2	Lab Submissions	Assignments	Class Participation	TEE
CO-1	Y		Y	Y	Y	Y
CO-2	Y	Y	Y	Y	Y	Y
CO-3		Y	Y	Y	Y	Y

5. Laboratory details

Tentative laves for this course are listed below. These labs may be revised during the semester.

Exp. No.	Week No.#	Programming Topic	Mapped CO
1.	1	Study of Computer Forensics and different types of cyber-crimes.	CO3
2.	2	Study of different forensics tools	CO2
3.	3	Study of Windows Registry and file system	CO1
4.	4	Email Header Analysis and Tracing Recent TTPs Used by threat actors	CO2
5.	5	Network Forensics detection and analysis of indicators of compromise (IoCs) over the network	CO2
6.	6	Revision	
7.	7	Browser Forensics	CO2
8.	8	Live system and Malware analysis Focus on Ransomware Wanna Cry	CO2
9.	9	Mobile Device Forensics	CO2
10.	10	Report writing	CO3
11.	11	Research paper presentation	CO1, CO2, CO3
12.	12	Research paper presentation	CO1, CO2, CO3
13.	13	Viva	CO1, CO2, CO3

14.	14	Viva	CO1, CO2, CO3
15.	15	Submissions	

6. Tutorial Plan - Nil

7. Course Material

References and Lab Manuals would be uploaded on LMS by faculty every week.

8. GenAI Usage

8.1 Pre-class Activity: Nil

8.2 In-Class Activity: Students may use Gen AI tools to explore cyber forensics topics.

Use of Gen AI tool for project report writing is STRICTLY Prohibited.

9. Academic Integrity Statement

Original work expected from students for all of the **assigned assessment work**. Copying in any form not acceptable and will invite strict disciplinary action. Evaluation of corresponding component will be affected proportionately in such cases. Plagiarism detection software will be used to check plagiarism wherever applicable. Academic integrity is expected from students in all components of course assessment.

*** - Only Teaching Method in the Instruction Plan for the course may vary for different faculty teaching the course, rest of the Course Policy Document will not change.**